Fibonacci with Dynamic Programming

Topic: DynamicProgramming File: dp1.pdf

PROBLEM

Compute the nth Fibonacci number, where F(0)=0, F(1)=1, and F(n)=F(n-1)+F(n-2).

IDEA

Store results instead of repeatedly solving the same smaller subproblems. A bottom-up loop only needs the two previous values.

PSEUDOCODE

```
if n <= 1: return n
a = 0; b = 1
for i = 2 to n:
    a, b = b, a + b
return b
```

EXAMPLE

F(5) = 5, from the sequence 0, 1, 1, 2, 3, 5.

COMPLEXITY

Time: O(n). Extra space: O(1) for the iterative version. Naive recursion takes exponential time.

TAKEAWAY

Dynamic programming avoids recomputing overlapping subproblems.